

JASON A. HERRMANN

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EDUCATION

UNIVERSITY OF CALIFORNIA, LOS ANGELES

Bachelor of Science in Electrical Engineering

Relevant Coursework: Circuit Design; Signal Processing; Digital Signal Processing

Los Angeles, CA

Sept. 2022 – Jun. 2026

PROJECTS

Noisemaker PCB Design

Jul. – Sept. 2026

- Reverse-engineered an 8-sound electronic noisemaker and designed a PIC12F1840-based PCB in KiCad; prototyped button-controlled PWM audio and embedded firmware on an STM32F405.

Visual Speech Recognition AI Model on Embedded Microcontroller

Jan. – Mar. 2026

- Designed and implemented a lightweight Visual Speech Recognition (lip-reading) AI system using low-resolution grayscale mouth-movement sequences, applying CNN-based spatiotemporal feature extraction.
- Optimized and deployed the model on an STM32 NUCLEO embedded microcontroller using INT8 quantization and model compression, achieving 82.5% classification accuracy with a 76KB of flash footprint and 20KB RAM usage.

Audio Signal Processing and Speech Classification

Nov. 2025

- Developed a Python speech-command classification pipeline using MFCC, LPC, and spectral features with XGBoost, evaluating preprocessing and feature-selection techniques for accuracy and robustness across clean, white-noise, and pink-noise datasets.

TECHNICAL SKILLS

Hardware/EDA: KiCad, LTSpice, Logisim Evolution

Embedded: STM32, STM32CubeIDE, STM32CubeProgrammer

Lab: Soldering, breadboarding, digital multimeter, PCB bring-up/debugging

Programming Languages: C, Python, MATLAB, JavaScript

ADDITIONAL EXPERIENCE

SUNSET CANYON RECREATION CENTER

Lifeguard

Los Angeles, CA

Jun. 2024 – Jun. 2026

SAN CLEMENTE AQUATICS CENTER

Lifeguard

San Clemente, CA

Jun. 2021 – Present

LEADERSHIP & ACTIVITIES

LAMBDA CHI ALPHA FRATERNITY

Design Chair

Los Angeles, CA

Sept. 2023 – Jun. 2026

- Designed merchandise and flyers using Adobe Photoshop for campus events.